PharmaGuide

SECOND EDITION · 2026

SUPPLEMENT SAFETY INTELLIGENCE BRIEF

Know what you're *actually* taking.

The pharmacist-reviewed field guide to labels, doses, hidden ingredients and interactions — built on primary research, written for real life.

60.2%	776	347%
of U.S. adults take a supplement	supplements FDA found hiding drugs	of labeled dose, in one tested product

REVIEWED BY
Laurie Pham, PharmD

12 SECTIONS · 21 CITED SOURCES

pharmaguide.io

INSIDE THIS BRIEF

Twelve sections. *One habit:* check before you take.

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HOW TO READ SEVERITY IN THIS BRIEF

CONTRAINDICATED	AVOID	CAUTION	MONITOR	INFORMATIONAL
Do not combine	Strong reason to separate or skip	Risk depends on context	Manage with timing or awareness	No action needed

Ratings describe general risk from published references. Your doses, conditions and medications can move an item up or down a tier.

THE STATE OF THE CABINET

A label tells you what's in one bottle. Not what happens with your *other* bottles.

Most people don't take one supplement. They take several, often alongside prescriptions. The risk usually isn't one ingredient — it's the combination, the dose that stacks, or the timing.

WHO TAKES SUPPLEMENTS

Any supplement, past 30 days — 60.2%



Two or more — 38.7%



CDC/NCHS Data Brief No. 561, 2026 · U.S. adults, 2021-2023

WHEN IT GOES WRONG

~23,000

ER visits a year linked to supplement adverse events

~2,150

hospitalizations a year

Geller et al., NEJM 2015 (PMID 26465986)

HIDDEN INGREDIENTS

776

supplements FDA found containing unapproved drugs, 2007–2016, from 146 companies

Tucker et al., JAMA Netw Open 2018

LIVER INJURY SHARE



of liver-injury cases at U.S. study centers tied to herbs & supplements, 2004–2013

Navarro et al., Hepatology 2014

> THE TAKEAWAY
>
> Many of these events may be avoidable by knowing exactly what you take, how much, and how it interacts. That is what this brief teaches.

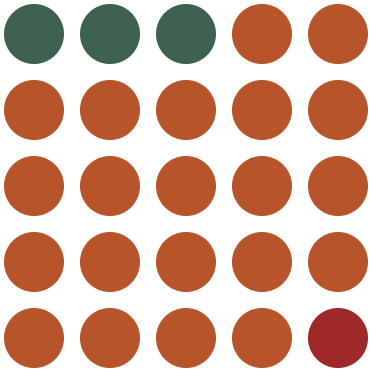
01

LABEL ACCURACY

What's on the label isn't always *what's in the bottle.*

In 2023, researchers bought 25 melatonin gummy brands and measured what was actually inside. It's the clearest recent picture of how far a supplement label can drift from reality.

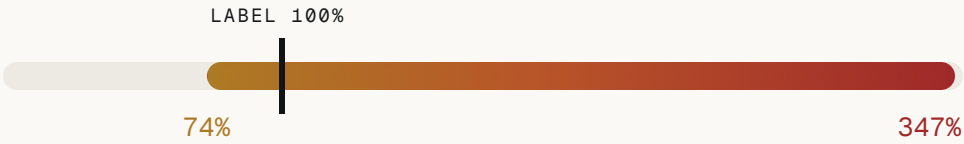
25 BRANDS TESTED



- 3 within ±10% of label
- 21 off by more than 10%
- 1 had no melatonin — only CBD

347%

The highest amount found, relative to the label. The range across products ran from **74% to 347%** of the stated dose — **1.3 to 13.1 mg** per serving.



Why the dose matters. The same researchers note that as little as 0.1–0.3 mg can lift blood melatonin into the normal nighttime range in young adults. Many gummies delivered far more.

The limits. The study tested gummies only, one sample per brand. It doesn't tell us about tablets or batch-to-batch variation — but it shows why "what the label says" is a claim, not a guarantee.

WHAT TO DO

Prefer products with an independent verification seal (Section 05). For sleep aids especially, start at the lowest available dose and ask your pharmacist.

Cohen PA, Avula B, Wang YH, Katragunta K, Khan I. Quantity of melatonin and CBD in melatonin gummies sold in the US. JAMA. 2023;329(16):1401-1402.

02

READ A LABEL IN 60 SECONDS

The front is marketing. The *back* is the truth.

Supplement Facts

2 Serving Size **2 Capsules**
Servings Per Container 30

	Amount Per Serving		%DV
1	Magnesium (as magnesium oxide)	400 mg	95%
4	Vitamin B6 (as pyridoxine HCl)	50 mg	2,941%
3	Calm Blend	450 mg	†
	L-theanine, ashwagandha root, valerian		

Other ingredients: rice flour, gelatin, magnesium stearate.

5 *"Clinically proven to cure stress."*

Illustrative label · not a real product

1 **The form, not just the milligrams**
"Magnesium (as magnesium oxide)" — the form sits in parentheses and changes how much is absorbed (Section 04).

2 **Serving size vs. what you'll take**
"2 capsules" means each capsule carries half the listed dose. Always divide.

3 **Blends hide the doses**
A "Calm Blend" lists ingredients but not amounts — so it can't be checked against your medications (Section 03).

4 **%DV far above 100%**
2,941% isn't a bonus. For some nutrients, it pushes you toward the upper limit (Section 06).

5 **Claims that sound medical**
"Cures" or "treats" a disease isn't legal language for a supplement. Treat it as a red flag about the brand.

03

THE PROPRIETARY BLEND TRAP

If the doses are *hidden*, safety can't be assessed.

A proprietary blend lists several ingredients under one total weight, without saying how much of each. It's legal — and common in sleep, energy and pre-workout products.

Sleep & Calm Blend · 1,200 mg — NOT ASSESSABLE

Magnesium	?
L-theanine	?
Valerian root	?
Melatonin	?
Herbal extracts	?

Fully disclosed formula — ASSESSABLE

Magnesium glycinate	200 mg
L-theanine	200 mg
Valerian root extract	300 mg
Melatonin	0.5 mg

01

Dose can't be confirmed

You can't tell if an active is at a meaningful amount — or mostly filler.

02

Interactions can't be checked

A pharmacist can't assess a dose they can't see against your medications.

03

Limits can't be tracked

Unknown amounts can't be added up against upper limits (Section 06).

WHAT TO DO

Prefer fully disclosed formulas, where every active ingredient lists its own amount. Transparency is the fastest signal that a brand expects scrutiny.

Fair note: a blend isn't automatically unsafe. It's unassessable — and for anyone on prescription medications, that alone is a reason to choose differently.

04

FORMS THAT ABSORB

Same milligrams. *Not* the same result.

The form decides how much is absorbed and how well it's tolerated. It hides in parentheses on the label.

NUTRIENT	OFTEN LESS ABSORBED / HARSHER	OFTEN BETTER ABSORBED / TOLERATED	WHY
Magnesium	Oxide	Citrate · Chloride · Lactate	NIH notes these forms absorb more completely than oxide in small studies. Glycinate is often chosen for being gentle on the stomach. Oxide may still suit some uses (e.g. constipation).
Calcium	Carbonate (no food)	Citrate	Carbonate needs stomach acid and food; citrate absorbs with or without — helpful if you take acid reducers.
Iron	Ferrous sulfate (GI upset)	Ferrous bisglycinate	Sulfate works but is hard on the gut for many. In trials — mostly in pregnant women — bisglycinate caused fewer stomach side effects than other iron salts.
Folate	—	Folic acid · L-methylfolate	Folic acid has established evidence for preventing neural-tube defects. L-methylfolate is an active form, but common MTHFR variants are not a reason to avoid folic acid.
Vitamin B12	—	Cyanocobalamin · Methylcobalamin	Both forms work for most people; methylcobalamin is an active form some clinicians and consumers prefer.
CoQ10	—	Ubiquinone · Ubiquinol	Small studies in older adults are mixed: one found ubiquinol raised CoQ10 more, another found no significant difference. The formulation matters too.
Zinc	Oxide	Citrate · Gluconate	In a controlled study, oxide was absorbed less than citrate or gluconate — and barely at all by some people.

Reality check: for many people a well-made standard form is fine. Form matters most when you have a specific reason — ask your pharmacist.

Sources: NIH ODS fact sheets [09] · Firoz 2001 [08] · CDC [14] · Wegmüller 2014 [15] · Fischer 2023 [16] · Zhang 2018 [17] · Pravst 2020 [18]

05

THIRD-PARTY SEALS

The FDA does *not* approve supplements before they're sold.

Unlike drugs, supplements rarely need pre-market proof of safety or label accuracy. An independent seal is the closest thing to outside verification.

- **USP Verified**
  Identity, potency, purity and dissolution; made under good manufacturing practices.
- **NSF Contents Certified**
  What's on the label is in the bottle; screened for contaminants.
- **NSF Certified for Sport**
  Everything above, plus screening for substances banned in sport.
- **Informed Sport / Informed Choice**
  Banned-substance tests; Sport checks every batch.
- **BSCG Certified Drug Free**
  Tests every product for banned substances; label and contaminants checked yearly.
- **ConsumerLab**
  Independent product testing and reviews (subscription).

A SEAL DOES TELL YOU

✓ (USP, NSF) Contents match the label
✓ (USP, NSF) Screened for contaminants
✓ (Sport seals) Tested for banned substances

A SEAL DOES NOT TELL YOU

✕ That it works for you
✕ That the dose is right for you
✕ That it's safe with your medications

WHAT TO DO

Treat a seal as a floor, not a guarantee. Verify it on the certifier's own website — seals can be copied onto packaging.

06

UPPER LIMITS & STACKING

The dose that *stacks* is the one people miss.

UL = Tolerable Upper Intake Level — the most of a nutrient you can take each day before the risk of side effects starts to rise.

EXAMPLE · VITAMIN D ACROSS THREE PRODUCTS — ILLUSTRATIVE



Each product looks reasonable alone. Together they exceed the adult upper limit by 75%.

NUTRIENT	UPPER LIMIT (UL) / DAY	WHAT GOES WRONG ABOVE IT
Vitamin D	4,000 IU (100 mcg)	Raises blood calcium; almost always from over-supplementing.
Calcium	2,000-2,500 mg	Food + supplements; limit set for kidney-stone risk. Can also cause constipation.
Magnesium (supplemental)	350 mg	Supplements only; above this, loose stools are common.
Zinc	40 mg	Chronic high intake lowers copper.
Iron	45 mg	Above this, stomach side effects become more likely. High-dose iron should be clinician-guided; very large overdoses can be toxic.
Vitamin A (preformed)	10,000 IU (3,000 mcg)	Liver toxicity; birth-defect risk in pregnancy.
Vitamin B6	100 mg	U.S. UL is 100 mg/day. In 2023, EFSA set 12 mg/day for adults, after finding neuropathy may occur at intakes as low as about 50 mg/day.
Niacin	35 mg (supplemental)	Flushing above this; high doses need medical supervision.

NIH Office of Dietary Supplements (ods.od.nih.gov) · ULs vary by age, sex and pregnancy; most include food and supplements.

07

INTERACTIONS PEOPLE MISS

The one nobody mentioned is often the one that *matters.*

Well-documented pairs. Not a complete list — a prompt to review yours with a pharmacist.

SUPPLEMENT	WITH	TIER	MECHANISM	WHAT TO DO
St. John's Wort	SSRIs / antidepressants	CONTRAINDICATED	Risk of serotonin syndrome; also speeds breakdown of many drugs, weakening them.	Don't combine.
Red yeast rice	Statins	AVOID	Contains a natural statin — doubling up raises muscle & liver risk.	Treat as a statin; don't add to one.
Vitamin K	Warfarin	CAUTION	Directly opposes the blood thinner; intake swings destabilize INR.	Keep intake consistent; tell your clinician.
Fish oil · Vitamin E	Blood thinners / antiplatelets	CAUTION	Possible added antiplatelet / bleeding effect, mainly at higher doses.	Review with your blood-thinner prescriber; flag before surgery.
Calcium · Magnesium · Iron	Levothyroxine	CAUTION	Bind the drug in the gut, reducing absorption — about 20–25% with calcium in one study.	Separate by at least 4 hours.
Calcium · Magnesium · Iron	Tetracycline & fluoroquinolone antibiotics	CAUTION	Minerals chelate the antibiotic, reducing its effect.	Separate doses by several hours.
Potassium	ACE inhibitors / ARBs	AVOID	Can push potassium too high (hyperkalemia).	Don't add potassium without supervision.

Don't stop a prescription on your own. Bring the combination to your pharmacist — the fix is usually timing or monitoring.

Sources: Zamfirescu 2011 [07] · NCCIH [12] · FDA labeling · NIH resources · severity is general.

08 HIDDEN DRUGS

Some supplements aren't supplements. They're *unlabeled drugs.*

Researchers analyzed a decade of FDA warnings about supplements found to contain unapproved pharmaceutical ingredients.

776

adulterated supplements identified by FDA, 2007–2016 · 146 companies

WHAT THEY WERE SOLD FOR



Muscle building 11.9% · other 1.8%

MARKETED FOR	MOST COMMON HIDDEN DRUG	SHARE OF THAT CATEGORY
Sexual enhancement	Sildenafil (the active drug in Viagra)	47.0%
Weight loss	Sibutramine (withdrawn from the U.S. market)	84.9%
Muscle building	Synthetic steroids or steroid-like ingredients	89.1%

20.2%
contained more than one unapproved drug

28
products named in repeat FDA warnings, months apart — warnings don't always end it

RED FLAGS ON THE PACKAGE OR LISTING

- ✕ Promises fast or dramatic results
- ✕ Words like guaranteed, miracle, or scientific breakthrough
- ✕ "Natural alternative" to a prescription drug
- ✕ Sold mainly online or shipped from overseas

WHAT TO DO

Be most skeptical in these three categories. Before buying, search the product on FDA's tainted-supplements and recall listings, and turn on FDA recall alerts.

Tucker J et al. Unapproved pharmaceutical ingredients included in dietary supplements associated with US FDA warnings. JAMA Netw Open. 2018;1(6):e183337. · Red flags: FDA health-fraud guidance [20].

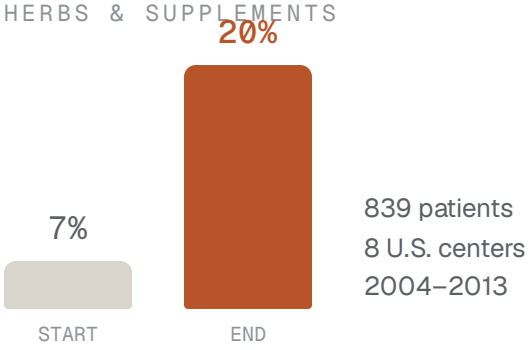
09

YOUR LIVER

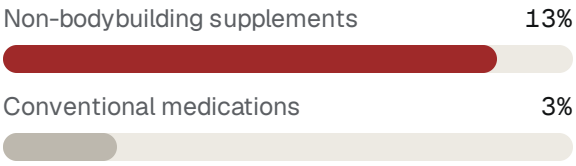
The organ that processes *everything* you take.

The U.S. Drug-Induced Liver Injury Network tracks liver damage from medications and supplements at specialist referral centers.

SHARE OF LIVER-INJURY CASES FROM HERBS & SUPPLEMENTS



DEATH OR LIVER TRANSPLANT



Non-bodybuilding products caused the more severe cases — mostly in middle-aged women.

COMMONLY IMPLICATED

Bodybuilding products with steroid-like ingredients · green tea extract [19] · multi-ingredient products

WARNING SIGNS — SEE A CLINICIAN

Yellowing skin or eyes · dark urine · persistent nausea · upper-right abdominal pain · unusual fatigue

LOWER YOUR RISK

Tell every clinician

List herbal products and powders, not just pills — they're the ones most often left off.

Fewer ingredients

Multi-ingredient "fat burner" and bodybuilding formulas are harder to trace when something goes wrong.

Stop and ask

If warning signs appear, stop the product and bring the bottle to your appointment.

Read this carefully: these are referral-center cases, not national rates — the authors caution it doesn't prove liver injury is rising population-wide. It does show supplements are a real, growing share of the cases specialists see.

Navarro VJ et al. Liver injury from herbals and dietary supplements in the U.S. Drug-Induced Liver Injury Network. Hepatology. 2014;60(4):1399-1408. · Warning signs: NIH LiverTox [19].

10

TIMING & SPACING

When you take it can matter as much as *what* you take.

A sample day for someone on thyroid medication who also takes common minerals. It's an example of the spacing logic — not a prescription.

WAKE

Thyroid medication (levothyroxine)

Empty stomach, water only. Wait 30–60 min before food.

BREAKFAST

Fat-soluble vitamins (A, D, E, K)

Take with a meal that has some fat. Keep iron and calcium away from thyroid meds (4+ hrs), including a multivitamin with minerals.

MIDDAY

Iron

Away from calcium and coffee/tea; vitamin C helps absorption.

DINNER

Calcium

Carbonate with food. Separate from iron; 4+ hrs after thyroid meds.

EVENING

Magnesium

Often better tolerated in the evening; split large doses.

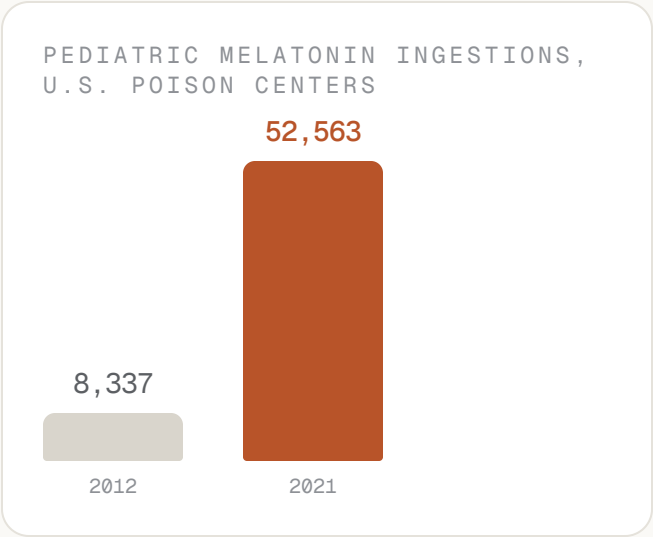
Metformin users: long-term use is linked to B12 depletion — worth periodic monitoring with your clinician.

High-dose zinc: can deplete copper over time — avoid taking it at the same time as iron or calcium.

11

KIDS & THE HOUSEHOLD

Gummies look like *candy*. Kids treat them that way.



+530%

rise in yearly pediatric melatonin ingestions reported to poison centers. Most involved young children, and most were unintentional.

HOUSEHOLD RULES THAT WORK

Store it like medicine
Up high, out of sight, child-resistant cap closed — every time.

Never call it candy
Kids copy adults. Take yours where they can't watch.

Ask before giving it to a child
Check with your pediatrician first, including sleep aids.

Count the gummies
Knowing how many were in the bottle helps if one goes missing.

Child swallowed too much? Call Poison Control right away. **1-800-222-1222**

Lelak K et al. Pediatric melatonin ingestions — United States, 2012-2021. MMWR Morb Mortal Wkly Rep. 2022;71:725-729. · Poison Help: HRSA [21].

12

THE CHECKLIST

Eight checks before you buy. *Or one scan.*

By hand, this takes a few minutes per bottle. Screenshot this page for the store — or let PharmaGuide help.

- [ ] **Every ingredient shows its amount** 01
  No "blend" that hides how much of each thing is inside.
- [ ] **It has a testing seal** 02
  USP or NSF on the label. Look it up on the seal's website to make sure it's real.
- [ ] **It uses a form your body absorbs well** 03
  Example: magnesium citrate instead of magnesium oxide. See page 7.
- [ ] **You're not doubling up** 04
  Add up the same vitamin across all your bottles. Stay under the daily limit on page 9.
- [ ] **It's safe with your medicines** 05
  See page 10 — and ask your pharmacist.
- [ ] **It's not a risky type of product** 06
  Be extra careful with sex-boost, weight-loss and muscle products.
- [ ] **The brand has no FDA warnings or recalls** 07
  Search the brand name plus "FDA warning" or "recall."
- [ ] **The promises sound realistic** 08
  No "cures," "miracle" or "works overnight."

PharmaGuide checks most of these for you.

Scan a bottle; add your meds for personal checks.

pharmaguide.io →

At home, that part's on you

Keep bottles up high, away from kids. Throw out anything expired.

Educational only · Not medical advice · Reviewed by Laurie Pham, PharmD

✦ TOOL · QUESTIONS FOR YOUR PHARMACIST

Ask your pharmacist. *Most people never do.*

Bring your bottles — or the worksheet on the next page — and ask these seven questions. It takes five minutes at the counter.

Q1	Is this supplement safe with everything else I take — including over-the-counter medicines?
Q2	Should I separate it from any of my medications? By how many hours?
Q3	Is this the right form and dose for me, given my age and conditions?
Q4	Does anything I take deplete a nutrient I should be monitoring?
Q5	Should I stop it before surgery or a procedure? How far ahead?
Q6	Is this product in a category with a history of hidden ingredients?
Q7	What side effects would mean I should stop and call you?

> PRO TIP
>
> Use one pharmacy for all prescriptions so their system can catch drug interactions — then tell them about your supplements, which their system usually can't see.

✦ TOOL · MY STACK WORKSHEET

Everything you take, *on one page.*

Include prescriptions, over-the-counter medicines, vitamins, herbs and powders. Print it, fill it in, bring it to every appointment.

PRODUCT / MEDICATION	DOSE	WHEN	WHY I TAKE IT	PRESCRIBED? Y/N

CONDITIONS & ALLERGIES

QUESTIONS TO ASK

SOURCES

Every claim, *traceable.*

01 Cohen PA, Avula B, Wang YH, Katragunta K, Khan I. Quantity of melatonin and CBD in melatonin gummies sold in the US. JAMA. 2023;329(16):1401–1402. doi:10.1001/jama.2023.2296

02 Lelak K, Vohra V, Neuman MI, Toce MS, Sethuraman U. Pediatric melatonin ingestions — United States, 2012–2021. MMWR. 2022;71:725–729. doi:10.15585/mmwr.mm7122a1

03 Tucker J, Fischer T, Upjohn L, Mazzera D, Kumar M. Unapproved pharmaceutical ingredients included in dietary supplements associated with US FDA warnings. JAMA Netw Open. 2018;1(6):e183337. doi:10.1001/jamanetworkopen.2018.3337

04 Navarro VJ, Barnhart H, Bonkovsky HL, et al. Liver injury from herbals and dietary supplements in the U.S. Drug-Induced Liver Injury Network. Hepatology. 2014;60(4):1399–1408. doi:10.1002/hep.27317

05 Geller AI, Shehab N, Weidle NJ, et al. Emergency department visits for adverse events related to dietary supplements. N Engl J Med. 2015;373:1531–1540. PMID 26465986.

06 National Center for Health Statistics. Dietary supplement use among adults: United States, August 2021–August 2023. NCHS Data Brief No. 561. 2026. cdc.gov/nchs/data/databriefs/db561.pdf

07 Zamfirescu I, Carlson HE. Absorption of levothyroxine when coadministered with various calcium formulations. Thyroid. 2011;21(5):483–486. doi:10.1089/thy.2010.0296

08 Firoz M, Graber M. Bioavailability of US commercial magnesium preparations. Magnes Res. 2001;14(4):257–262. PMID 11794633.

09 NIH Office of Dietary Supplements. Health professional fact sheets: vitamin D, calcium, magnesium, zinc, iron, folate, vitamin B12, vitamin A, vitamin B6, niacin. ods.od.nih.gov

10 EFSA NDA Panel. Tolerable upper intake level for vitamin B6. EFSA J. 2023;21(5):8006. doi:10.2903/j.efsa.2023.8006

11 U.S. Food and Drug Administration. Tainted products marketed as dietary supplements; supplement recalls and safety alerts. fda.gov

12 NIH National Center for Complementary and Integrative Health. St. John's wort and other herb–drug interaction fact sheets. nccih.nih.gov

13 Manufacturer and certifier program descriptions: USP, NSF International, LGC Informed Sport, BSCG, ConsumerLab.

14 Centers for Disease Control and Prevention. MTHFR gene variant and folic acid facts. cdc.gov/folic-acid/data-research/mthfr

15 Wegmüller R, Tay F, Zeder C, Brnic M, Hurrell RF. Zinc absorption by young adults from supplemental zinc citrate is comparable with that from zinc gluconate and higher than from zinc oxide. J Nutr. 2014;144(2):132–136. PMID 24259556.

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17 Zhang Y, Liu J, Chen XQ, Chen CYO. Ubiquinol is superior to ubiquinone to enhance coenzyme Q10 status in older men. Food Funct. 2018;9(11):5653–5659. doi:10.1039/c8fo00971f

18 Pravst I, Rodríguez Aguilera JC, Cortes Rodriguez AB, et al. Comparative bioavailability of different coenzyme Q10 formulations in healthy elderly individuals. Nutrients. 2020;12(3):784. ncbi.nlm.nih.gov/pmc/articles/PMC7146408

19 National Institutes of Health. LiverTox: clinical and research information on drug-induced liver injury. ncbi.nlm.nih.gov/books/NBK547852

20 U.S. Food and Drug Administration. Health fraud scams: consumer guidance on spotting fraudulent health products. fda.gov

21 Health Resources and Services Administration. Poison Help line, 1-800-222-1222. poisonhelp.hrsa.gov

This brief is educational and was reviewed for accuracy and clarity by Laurie Pham, PharmD. It is not a substitute for individual medical advice.

PharmaGuide

THIS BRIEF IS THE MANUAL VERSION

Check your *whole stack* in seconds.

Sourced from NIH's 200,000+ supplement labels, PharmaGuide checks your stack against your medications and conditions — with the reasoning and sources behind every flag. Core checks run privately on your device.

- ✓ Drug–supplement interactions
- ✓ Supplement–supplement interactions
- ✓ Recalled & banned ingredients
- ✓ Duplicate ingredients
- ✓ Upper-limit tracking
- ✓ Condition-specific risks
- ✓ Medication nutrient depletion
- ✓ Third-party testing status

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